Energy Management (EMM)

General information

In the vehicle, HV energy management regulates the provision of electrical energy and thus represents the direct connection between the HV energy storage system and the installed HV electrical components. The energy provided must always be limited to component-compatible levels. Communication between EMM and the HV components takes place via bus systems.

The following system functions can be important here:

- Operational release and limitation of the current and voltage limits of all HV components
- Operational activation and deactivation of the HV battery
- Current limitation in the HV wiring harness
- Coordination of HV battery charging processes
- Calculation of electric range
- LV wiring system support
- Manage departure and loading times

In addition, tasks are derived from functional safety for energy management that require increased attention during implementation and validation (ASIL requirements/tests).

The EMM is a very comprehensive system that is linked to many other systems. For this reason, testing requires a very good understanding of the overall system.

Software Management (SMM)

General information

The software management (SMM) coordinates and monitors the transition between different operating states (electric driving, hybrid driving, etc.). This means that SMM is closely linked to drive management (VMM). In addition to the software-side process monitoring, the SMM takes over the definition of substitute reactions, for example, to be able to avoid error states of the power electronics by targeted regulation.

The following system functionalities are important here:

- Defining target and prohibition states
- Higher-level monitoring concepts of the control unit
- Plausibility check of driver attendance
- Shift-by-Wire Plausibility Check
- Interface for diagnostic services and variant coding
- Monitoring of sensor data

Software management covers an extremely security-relevant subject area. In addition to an understanding of the driving system, testing also requires knowledge of data processing.

Thermal management (TMM)

General information

In hybrid and electric vehicles, thermal management (TMM) regulates the temperatures of the HV systems and the interior. Since HV components, such as the battery or the electric motors, only function optimally in a certain temperature range, they must be temperature-controlled. The TMM monitors the temperatures of these components and climatizes them in the event of deviations.

The following system functionalities can be important here:

- Temperature control of HV systems (battery, electric motors)
- Air conditioning of the vehicle interior
- Error monitoring of temperature control systems
- Optimization of battery performance and range through temperature monitoring
- Optimization of charging time through efficient cooling of the battery
- Increased service life by minimizing thermal stress

The TMM covers a subject area that is linked to different systems. For this reason, testing requires a good understanding of the overall system

Drive management (VMM)

General information

In hybrid and electric vehicles, drive management (VMM) primarily regulates the power requirements for the corresponding drive components and thus the driving behavior. When starting up, creeping or during dynamic driving, performance specifications for the electric powertrain must be coordinated within situational system boundaries through communication with relevant interfaces. The VMM thus represents the "planning centre", which acts as a link between energy supply (energy management) on the one hand and performance of the powertrain on the other.

The following system functionalities can be important here:

- Vehicle's operating strategy (recuperation, boost)
- Drive condition depending on energy management and drive components
- Fault state of involved components
- Starting and stopping the powertrain
- Switching on the combustion engine (hybrids) in the event of critical energy supply or increased power demand
- Representation of mileage
- Driving programs (Sport, Eco, etc.)

The VMM covers a wide range of topics, which are divided into numerous system function clusters. For this reason, testing requires a broad-based understanding of the overall system, which is linked to energy supply and power electronics.

Testing Requirements

In the case of platform software, this is used in various vehicles (PHEVs and BEVs) and networking architectures. To do justice to all projects, we rely on intelligent and expandable variant handling.

If the appropriate functional frameworks are in place, abnormalities are analyzed down to their origin. Furthermore, the consistency between requirements and implementation is checked.

Networking Testing

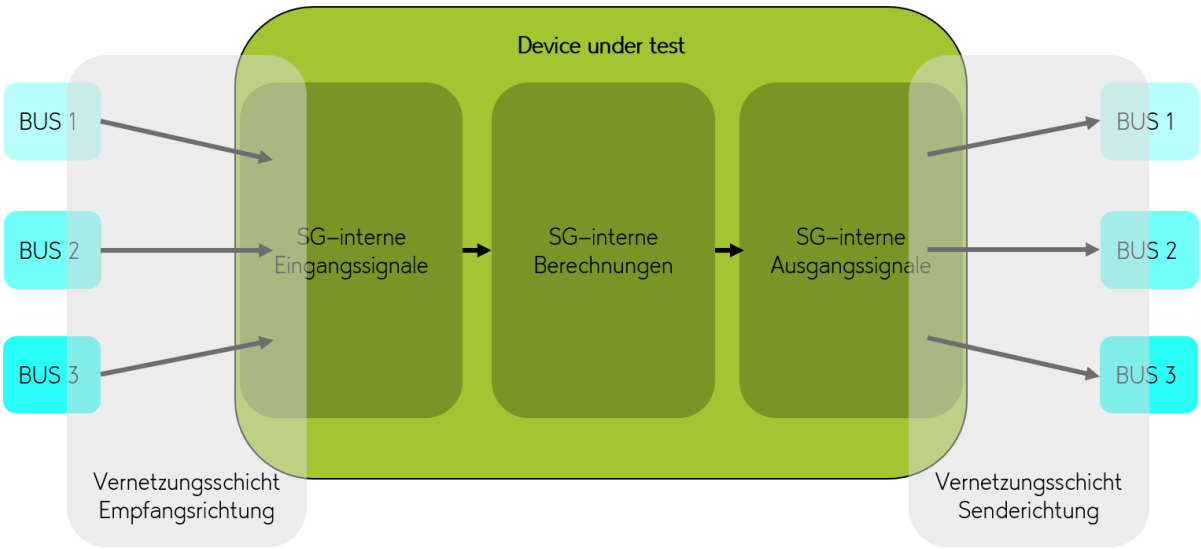
General information

In addition to the functional tests, we also carry out networking tests for the Central Driving and Charging Controller (CDCC = Device under test) on the HiL test bench. The networking tests include the following criteria:

- Verification of the receiving and transmitting functionality
- Comparison of implementation against the requirements
- Comparison of value ranges and scales between BUS signal and SG internal variable
- Limit value analysis and verification of correct limitation (keyword: overflow)
- Testing of signal and message failures

Testing Requirements

Special prerequisites are required for the direction of transmission and reception. These are shown by the following diagram:



Input Networking

In the input networking tests, BUS signals from other (→ not the device under test) control units simulated on the HiL are directly overwritten and the reaction in the networking layer is checked at least up to the first SG-internal variable. Here, both the valid value range and the replacement value behavior are validated in the event of signal or message failure.

Output Networking

To be able to test the entire value range of the internal output signals, the underlying SW module is switched off. By means of the INCA connection, the entire value range of the output signals can be manipulated via adaptive adjustment variables, while the SW module is not calculated. The BUS signal is used to check the reaction to the adjustment of the internal output variable.
This procedure also ensures that signal areas that occur very rarely in normal operation can also be checked.